Paper 3D: Decision Mathematics 1*

Topics	What students need to learn:		
	Content		Guidance
1 **Algorithms and graph theory**	1.1	**The general ideas of algorithms and the implementation of an algorithm given by a flow chart or text.**	**The meaning of the order of an algorithm is expected. Students will be expected to determine the order of a given algorithm and the order of standard network problems.**
	1.2	**Bin packing, bubble sort and quick sort.**	**When using the quick sort algorithm, the pivot should be chosen as the middle item of the list.**
	1.3	**Use of the order of the nodes to determine whether a graph is Eulerian, semi-Eulerian or neither.**	**Students will be expected to be familiar with the following types of graphs: complete (including K notation), planar and isomorphic.**
	1.4	The planarity algorithm for planar graphs.	Students will be expected to be familiar with the term 'Hamiltonian cycle'.